Oleksandr Gudzenko

Senior Python & Robotics Software Engineer

ROS 2 · Autonomous Navigation · Humanoid & Service Robots · AI Agents

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PROFESSIONAL SUMMARY

Senior Python and robotics software engineer with 10+ years of experience developing, integrating and deploying software for physical robots. Hands-on experience spans service, reception and hospitality robots—including Pepper, NAO, Sanbot, OrionStar and Reeman—and modern humanoid platforms from Unitree, Booster Robotics and Agibot.

Experienced in ROS 2, autonomous navigation, SLAM, localization, simulation, robot communication and vendor SDK integration. Build complete robotics products across Python backend services, web, desktop and mobile operator applications, as well as AI agents, voice systems and computer-vision components.

Full-Stack Robotics Developer and Technical Lead managing five software developers and one QA engineer. Responsible for architecture, hands-on implementation, technical planning, code quality, mentoring, recruitment and customer delivery.

TECHNICAL SKILLS

Robotics
ROS 2, Nav2, SLAM Toolbox, AMCL, localization, path planning, obstacle avoidance, costmaps, TF, URDF/Xacro, SDF, lidar, IMU and odometry

Simulation & ROS
Gazebo Harmonic, RViz, ros_gz, rclpy, rclcpp, topics, services, actions, lifecycle nodes, QoS, launch, rosbag and colcon

AI & perception
AI agents, tool calling, RAG, OpenAI, PydanticAI, ElevenLabs, TTS, OpenCV, YOLO and PyTorch

Programming
Python, asyncio, C++, TypeScript, JavaScript, Dart and Java

Backend & communication
FastAPI, Django, Flask, REST, WebSocket, Socket.IO, microservices and event-driven systems

Applications & infrastructure
React, Flutter, Tauri, Docker, CI/CD, Linux, PostgreSQL, pgvector, Redis, RabbitMQ, Kafka and Celery

PROFESSIONAL EXPERIENCE

Delta Robots (formerly WeegreeOne), Poland

Full-Stack Robotics Developer / Technical Lead · Jul 2016 – Present

- Lead five software developers and one QA engineer while remaining actively involved in architecture and implementation.
- Design Python services and reusable integration layers for heterogeneous physical robots, vendor SDKs and communication protocols.
- Integrate, configure, deploy and diagnose software directly on service robots and modern humanoid platforms.
- Develop robot-control, navigation, speech, animation, camera, multimedia, monitoring and AI-conversation capabilities.
- Build operator applications with Flutter, React and TypeScript for Android, iOS, Windows, Linux and web.
- Own technical planning, estimation, code reviews, quality control, mentoring, recruitment and release coordination.
- Work directly with customers from discovery and product planning through deployment, training and production support.

Wyrażam zgodę na przetwarzanie moich danych osobowych dla potrzeb niezbędnych do realizacji procesu rekrutacji zgodnie z Rozporządzeniem Parlamentu Europejskiego i Rady (UE) 2016/679 z dnia 27 kwietnia 2016 r. w sprawie ochrony osób fizycznych w związku z przetwarzaniem danych osobowych i w sprawie swobodnego przepływu takich danych oraz uchylenia dyrektywy 95/46/WE (RODO).

SELECTED COMMERCIAL PROJECTS AT DELTA ROBOTS

Multi-Platform Robot Control and Scenario Orchestration System

Principal engineer for a commercial ecosystem used to control and orchestrate heterogeneous service and humanoid robots. The system combines a visual scenario designer with a Flutter application for direct remote operation and has been delivered to multiple customers.

- Designed common interfaces for robots with different capabilities, SDKs and communication protocols.
- Implemented Python services for movement, navigation, speech, AI conversations, audio, animations, camera access and device monitoring.
- Designed an event-driven runtime for visual behavior graphs with parallel execution, validation, cancellation and error handling.
- Integrated physical robots through vendor SDKs, WebSocket, Socket.IO and REST APIs.
- Supported platforms from Pepper, NAO, Sanbot, OrionStar and Reeman through newer humanoids from Unitree, Booster Robotics and Agibot.

Technologies: Python, Flutter, React, TypeScript, WebSocket, Socket.IO, REST, OpenAI, ElevenLabs, Docker and robot vendor SDKs

AI Reception Assistant

Developed a product line that evolved from physical reception robots into an AI-powered digital assistant for information kiosks and public spaces.

- Built multilingual interactions for welcoming visitors, presenting information, notifying employees and conducting safety training.
- Developed Python backend services and React-based administration interfaces.
- Integrated realtime speech recognition, TTS, AI conversations and knowledge-base-driven responses.
- Supported physical service robots, virtual avatars and standalone kiosk deployments.

Technologies: Python, Django, FastAPI, Flask, React, OpenAI, ElevenLabs, Google Speech Recognition, Docker and REST APIs

AI Agent and RAG Platform for Robots

Designed a Python platform connecting heterogeneous robots with configurable AI agents, tools and domain-specific knowledge bases.

- Implemented structured tool calling, multi-format document ingestion and configurable RAG pipelines.
- Built vector, full-text and hybrid retrieval using local embeddings and PostgreSQL/pgvector.
- Designed reusable interfaces for exposing robot capabilities to AI agents.

Technologies: Python, PydanticAI, OpenAI Responses API, PostgreSQL, pgvector, FastEmbed, SQLAlchemy, Docker Compose and Pytest

Other Robotics and Automation Projects

- Hospital robots for navigation, reception, temperature screening, rehabilitation and child entertainment.
- Natural-language industrial assistant for production and warehouse information.
- ABB YuMi robotic bartender with tablet control, RAPID programming, sensors and custom electronics.
- Automated recruitment voicebot using Django, React, Twilio, Celery and Redis.
- Business-process automation using n8n, webhooks, queues and third-party integrations.

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PERSONAL ROBOTICS AND COMPUTER VISION PROJECTS

Cargo Bot — Personal ROS 2 Project

Designed an autonomous warehouse robot simulation as a public ROS 2 R&D project. The workspace contains seven ROS packages covering robot modeling, control, custom interfaces, simulation, SLAM, localization, autonomous navigation and automated verification.

- Created a differential-drive cargo robot and manipulator using URDF/Xacro.
- Implemented Python ROS 2 nodes for motion, odometry, joint states and manipulator actions.
- Built collision-enabled warehouse and multi-room environments in Gazebo Harmonic.
- Integrated lidar, wheel odometry and IMU data with multiple simulated sensor profiles.
- Implemented SLAM Toolbox mapping and AMCL localization.
- Configured Nav2 with Smac planning, Regulated Pure Pursuit control and behavior trees.
- Implemented dynamic-obstacle detection, replanning, persistent obstacle memory and collision-monitor safety zones.
- Developed a custom C++ Nav2 costmap plugin and end-to-end navigation tests.

Technologies: Python, C++, ROS 2 Jazzy, Nav2, SLAM Toolbox, AMCL, Gazebo Harmonic, RViz, ros_gz, URDF/Xacro and Pytest
Repository: github.com/Gudzenko/robotics_playground

YOLO Object Detection

Built a computer-vision project for training and applying custom YOLO models to chicken and wildlife detection in images and video.

- Implemented dataset acquisition and validation, GPU training, image/video inference and result generation.
- Added OpenCV and FFmpeg utilities for preparing and processing test media.

Technologies: Python, Ultralytics YOLO, PyTorch, CUDA, OpenCV and FFmpeg
Repository: github.com/Gudzenko/ImageTraining

ACADEMIC EXPERIENCE

Opole University of Technology

PhD research in Automation and Robotics

Research on control methods for an asymmetrical quadcopter: mathematical modeling, controller design, disturbance analysis, assembly, calibration and flight control. Author and co-author of related publications.

Kharkiv University of Technology

Research and teaching · 2010–2014

Created and taught mathematics and computer-science courses and developed research software using MATLAB, C++, Qt and Delphi.

EDUCATION

PhD studies, Automation and Robotics
Opole University of Technology

MSc, Applied Mathematics
Kharkiv University of Technology

BEng, Applied Mathematics
Kharkiv University of Technology

LANGUAGES

Polish — good working proficiency
English — intermediate
Russian — fluent
Chinese — basic

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